



Superficial cutaneous fungal infections

Primary care block/dermatology

Dr.fenik jabar yawar
Board certified dermatologist



Learning outcomes

- To know the different types of superficial fungal infections of skin
- To be familiar with the different types of fungi causing them
- To be able to diagnose those infections
- To know the different treatment strategies of them



Content

- Dermatophytosis *Tinea*
- candidiasis
- Pityrosporum infection
- Piedra
- Tinea nigra



Dermatophytosis

Three important types:

1. Trichophyton - Skin, hair, nail. *Trichophyton* *causality*
2. Epidermophyton - Skin, nail. *Epidermophyton*
3. Microsporum - Skin, hair. *Microsporum*

- Virulence factors of dermatophytes are **arthroconidium (asexual spores)**, **enzymes (keratinase, protease, lipase)**, and **mannan** of the fungal cell wall.

cut skin to enter
anti-inflammatory prevent antibody response

that survive for long time in soil, animal, human



Classification

→ from animal (more severe) → more inflammatory
more scoring

→ from enviro

→ from human

Zoophilic	Geophilic	Anthropophilic
<i>Trichophyton interdigitale</i> (former <i>T. mentagrophytes</i>)	<i>Microsporum</i> <i>gypseum</i>	<i>Epidermophyton</i> <i>floccosum</i>
<i>Trichophyton verrucosum</i>	<i>Microsporum</i> <i>ferrugineum</i>	<i>Trichophyton rubrum</i>
<i>Microsporum canis</i>	<i>Microsporum</i> <i>langeronii</i>	<i>Trichophyton</i> <i>raubitschekii</i>
		<i>Trichophyton</i> <i>violaceum</i>

SR



Clinical types

- Tinea corporis - ringworm infection of the body ^(body) ^{annular}
- Tinea pedis - ringworm infection of the foot ^{foot}
- Tinea cruris - ringworm infection of the groin ^(genital area, pubic)
- Tinea unguium - ringworm infection of the nails ^{nail}
- Tinea capitis - ringworm infection of the scalp ^{hair}
- Tinea manuum - ringworm infection of the hand ^{hand}
- Tinea barbae - ringworm infection of the beard ^{beard}



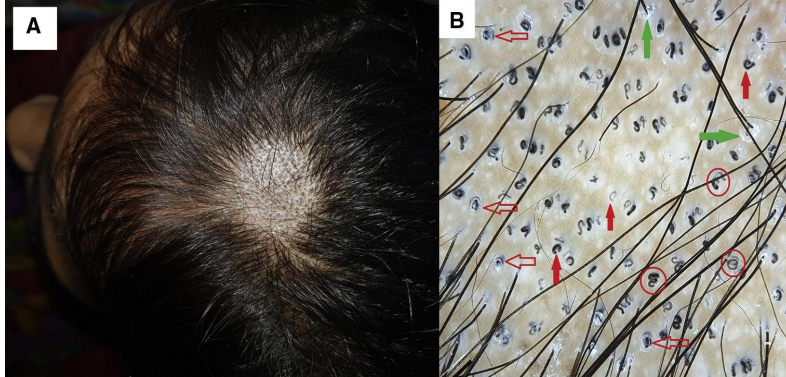
Presentation

Skin: Circular, dry, erythematous, ^{usually zoophilic} scaly, itchy lesions.
Hair: inflammatory (keroin ,favus) , non inflammatory (black dot ,gray patch).
Nail: Thickened, deformed, friable, discolored nails, sub-ungual hyperkeratosis.



Tinea capitis

flor. inflammatory



Inflammatory, scarring





Candidiasis

- Candida albicans , C. Tropical , C.glabrata
- Mainly affecting genitalia , nails , oral mucosa
- Causing oral thrush , angular cheilitis, paronychia , interdigital candidiasis , intertrigo , vulvovaginitis.

inflammation of
skin around nail

→ skin folds (oral fold)





Pityrosporum infection

- Pityriasis versicolor :

The yeast that causes tinea versicolor, *Malassezia*, grows on normal, healthy skin. But the following things can trigger an overgrowth that causes the infection:

- Oily skin
- Living in a hot climate
- Sweating a lot
- Hormonal changes *like pregnancy*
- A weakened immune system.

Because the yeast grows naturally on our skin, tinea versicolor isn't contagious.



Clinical features

Tinea versicolor signs and symptoms include:

- Patches of skin discoloration, usually on the back, chest, neck and upper arms, which may appear lighter or darker than usual *→ versicolor = change color*
- Mild itching
- Scaling

oily places





Tinea nigra

- Tinea nigra is a fungal infection of the skin of the palm or sole presenting as persistent brown or black patches.
- Tinea nigra is most common in tropical regions and often infects those with a tendency to excessive sweating.
- Tinea nigra is due to infection with a brown fungi, *Hortaea werneckii*. This mould usually inhabits soil.





Piedra

hair beading areas
in hot climate areas
weaken hair shaft to fracture and loss

- Piedra is a superficial fungal infection of the hair shaft seen most commonly in tropical climates. It presents in two distinct clinical varieties, black piedra and white piedra, caused by *Piedraia hortae* and Trichosporon species, respectively.
- Black piedra presents as asymptomatic, microscopic to 1 mm or larger, dark, firmly adherent, concretions on hair shafts of the scalp or, less commonly, the beard. White piedra also presents as asymptomatic concretions or nodules on the hair shafts, although these are lighter in color (white to light brown) and can be easily detached, unlike those of black piedra. White piedra involves facial and genital hair more often than scalp hair. In both forms, affected hair shafts may be weakened and fracture easily.

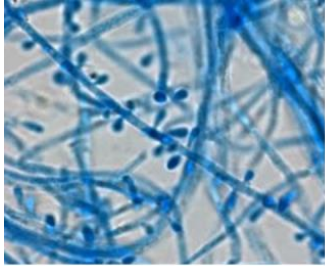




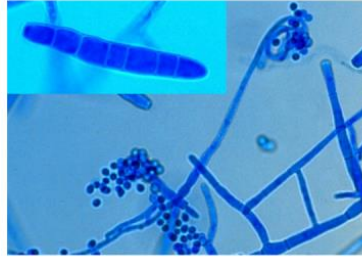
Diagnosis

- Wood's light examination *→ Vesicular doesn't change color*
- Dermoscopy
- 10% KOH
- culture (SDA)

Microscopic features



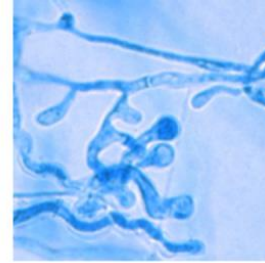
T. rubrum



T. mentagrophytes



T. tonsurans



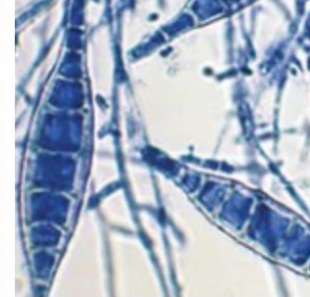
T. schoenleinii



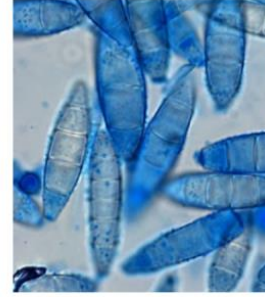
T. violaceum



M. audouinii



M. canis



M. gypseum

Versicolor

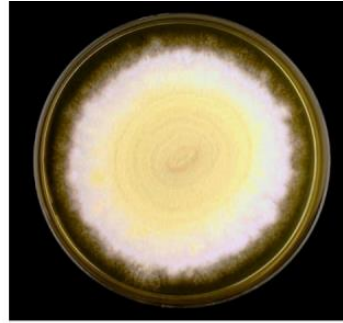
spaghetti and meatballs } *have oil place*
 even olive oil added to agar to produce more optimal growth



Macroscopic feature on SDA culture



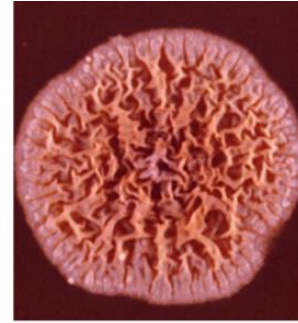
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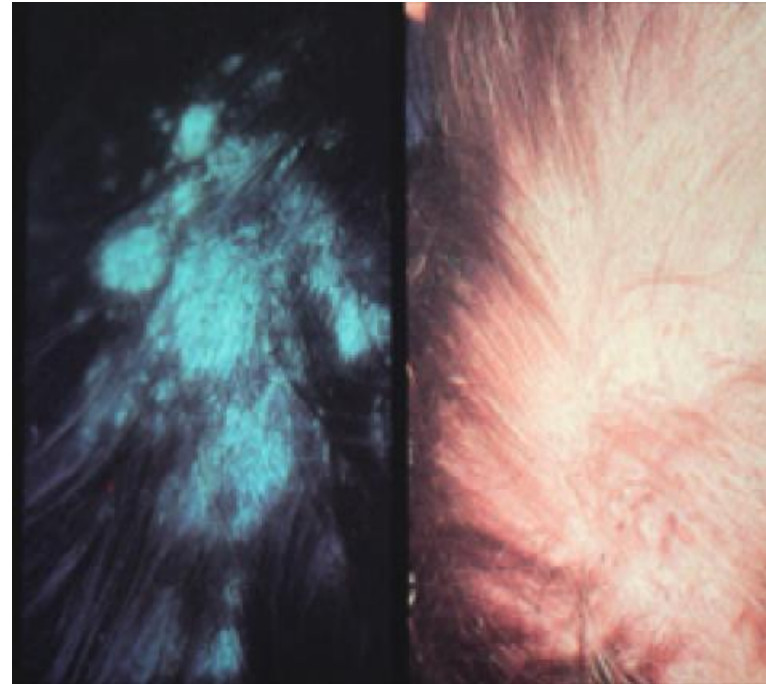
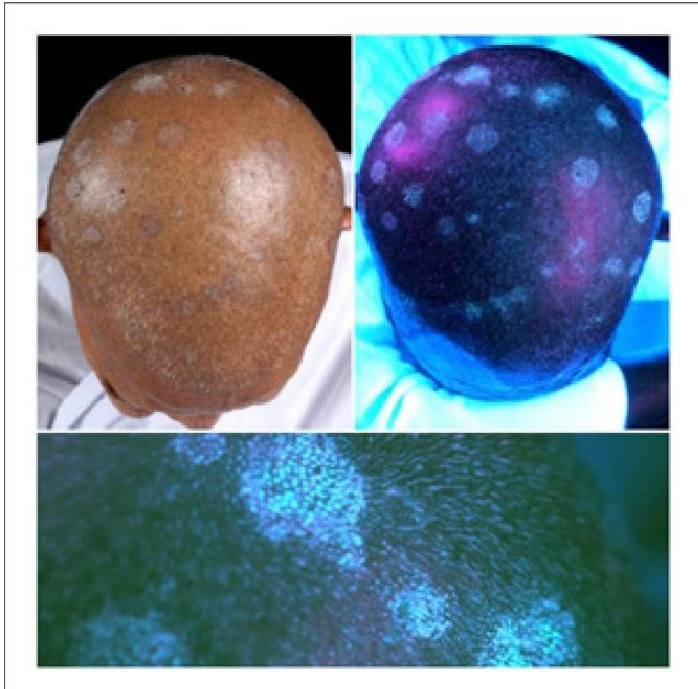
M. canis



M. gypseum



Wood's light





Treatment

- Topical anifungals : clotrimazole , imidazole , ketokonazol
- Systemic antifungals : terbinafine , itraconazole , griseofulvin , fluconazole
- maintaining hygiene
- trimming hair